Quasars

By the late 1950s, the Americans were competitive. Radio telescopes at Greenbank were coming into operation, and at Caltech, Tom Mathews, Per Eugen Maltby, and Alan Moffett on the new Owens Valley radio interferometer were working hand in hand with Baade, Greenstein, and others on the Palomar 5-meter optical telescope to discover and study large numbers of radio galaxies.

In 1960 this effort brought another surprise: Tom Mathews at Caltech received word from Henry Palmer that, according to Jodrell Bank measurements, a radio source named 3C48 (the 48th source in the third version of a catalog constructed by Ryle's group at Cambridge) was extremely small, no more than 1 arc second in diameter (1/2,000 of the angular size of the Sun). So tiny a source would be something quite new. However, Palmer and his Jodrell Bank colleagues could not provide a tight error box for the source's location. Mathews, in exquisitely beautiful work with Caltech's new radio interferometer, produced an error box just 5 seconds of arc in size, and gave it to Allan Sandage, an optical astronomer at the Carnegie Institution in Pasadena. On his next observing run on the 5-meter optical telescope, Sandage took a photograph centered on Mathews's error box and found, to his great surprise, not a galaxy, but a single, blue point of light; it looked like a star. "I took a spectrum the next night and it was the weirdest spectrum I'd ever seen," Sandage recalls. The wavelengths of the spectral lines were not at all like those of stars or of any hot gas ever manufactured on Earth; they were unlike anything ever before encountered by astronomers or physicists. Sandage could not make any sense at all out of this weird object.

Over the next two years a half-dozen similar objects were discovered by the same route, each as puzzling as 3C48. All the optical astronomers at Caltech and Carnegie began photographing them, taking spectra, struggling to understand their nature. The answer should have been obvious, but it was not. A mental block held sway. These weird objects looked so much like stars that the astronomers kept trying to interpret them as a type of star in our own galaxy that had never before been seen, but the interpretations were horrendously contorted, not really believable.